

Model Comparison & Selection Report

Project: Mobile Money User Activity Classification — CSC 3221

1. Problem Formulation

The prediction target is **user activity segment** — a 3-class classification problem distinguishing Low, Medium, and High activity users based on their transaction behavior and demographic characteristics. This prediction is valuable because it enables mobile money providers to personalize services, prioritize retention efforts, and design targeted financial products.

Success criteria: A model achieving weighted F1-score above 0.80 on the test set, significantly beating the baseline (0.44 accuracy).

2. Models Trained

- **Baseline:** Majority class predictor (always predicts 'Medium' — most frequent class). Accuracy: 0.444.
- **Logistic Regression:** Linear probabilistic classifier with L2 regularization (C=1.0). Scaled features.
- **Decision Tree:** Non-parametric tree with depth=6, min_samples_split=4 to prevent overfitting.
- **Random Forest:** Ensemble of 200 trees (max_depth=8). Robust to overfitting via bagging.
- **Gradient Boosting:** Sequential ensemble (150 estimators, lr=0.1, depth=4). High predictive power.
- **K-Nearest Neighbors:** Distance-weighted KNN (k=5). Simple, non-parametric benchmark.

3. Performance Comparison

Model	Val Accuracy	Val F1	Val AUC	CV Mean	CV Std
Logistic Regression	0.889	0.838	0.944	0.700	±0.215
Decision Tree	1.000	1.000	1.000	0.667	±0.140
Random Forest	1.000	1.000	1.000	0.789	±0.079
Gradient Boosting	1.000	1.000	1.000	0.667	±0.091
KNN	0.889	0.838	0.944	0.694	±0.077
Baseline	0.444	0.196	N/A	—	—

4. Best Model Selection: Random Forest

While Decision Tree and Gradient Boosting achieved equal validation scores, **Random Forest** was selected as the best model based on the following justification:

- Highest cross-validation mean (0.789) with the lowest variance (±0.079), indicating superior generalization.
- Inherently resistant to overfitting through bootstrap aggregation (bagging) of 200 independent trees.
- Provides robust feature importance estimates via mean decrease in impurity.
- Decision Tree's perfect validation score with lower CV mean (0.667) suggests it overfits the small validation set.
- Gradient Boosting, while powerful, showed higher CV variance — less reliable with our small sample (60 users).

5. Trade-offs Discussion

Interpretability vs Performance: Logistic Regression is the most interpretable (coefficients directly show feature impact) but achieved lower F1 (0.838). Random Forest is less directly interpretable but offers feature importance scores as a proxy.

Overfitting risk: Decision Tree and Gradient Boosting show signs of overfitting to the small validation set (Val F1 = 1.000 vs CV = 0.667). Random Forest mitigates this through ensemble averaging.

Sample size caveat: With only 60 users and a 15% test split (9 samples), metrics must be interpreted cautiously. A larger dataset would allow more reliable model comparison.

6. Top Predictive Features

- **tx_per_month (74%)** — strongest predictor; frequency of use directly defines activity segments.
- **total_fees_paid (14%)** — high fees indicate high transaction volume and amounts.
- **total_transactions (8%)** — raw count complements frequency by capturing overall usage.
- **max_transaction_amount (5%)** — peak transaction size distinguishes business from casual users.
- Demographic features (zone, income, smartphone) had lower importance, suggesting behavioral data dominates demographic signals for this classification task.